Poster Nº P027-M

Data integration analysis for profiling host-microbiome interactions in non-responsive Celiac disease patients

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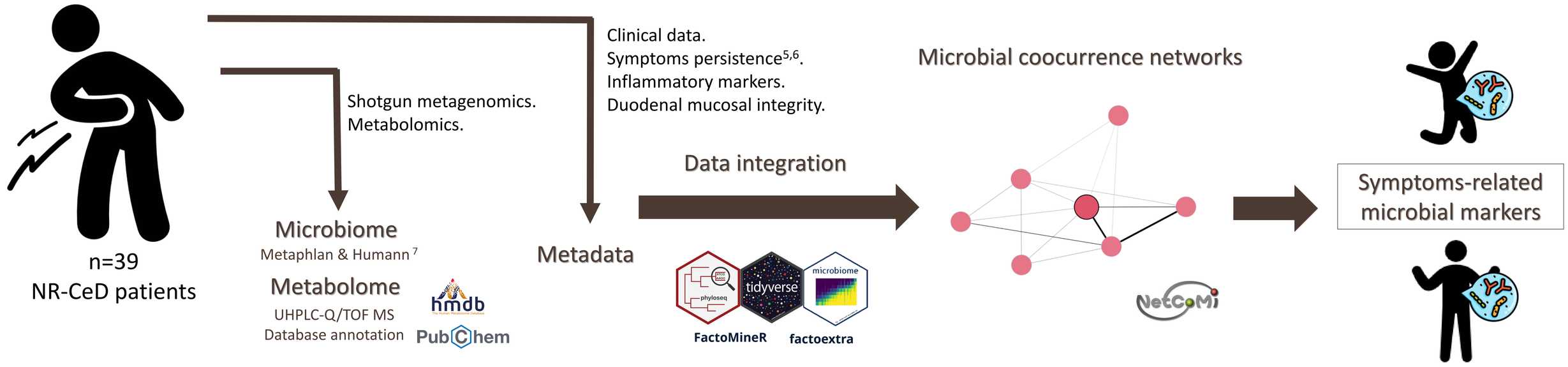
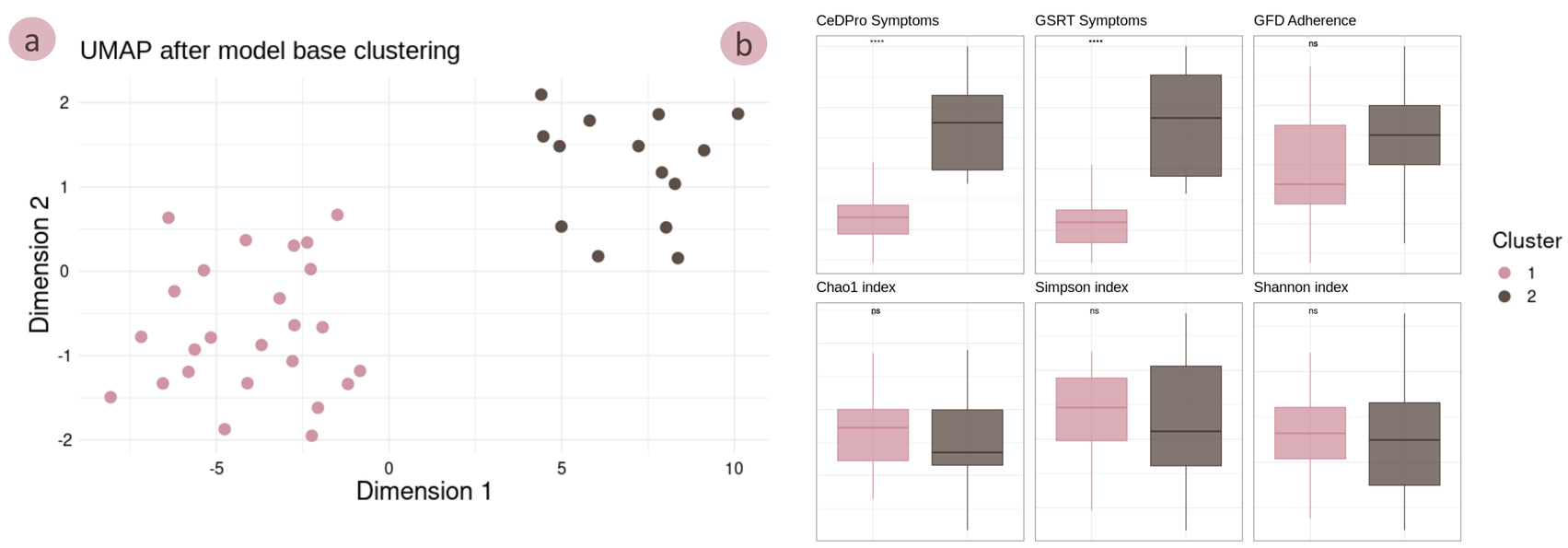
Background & methods

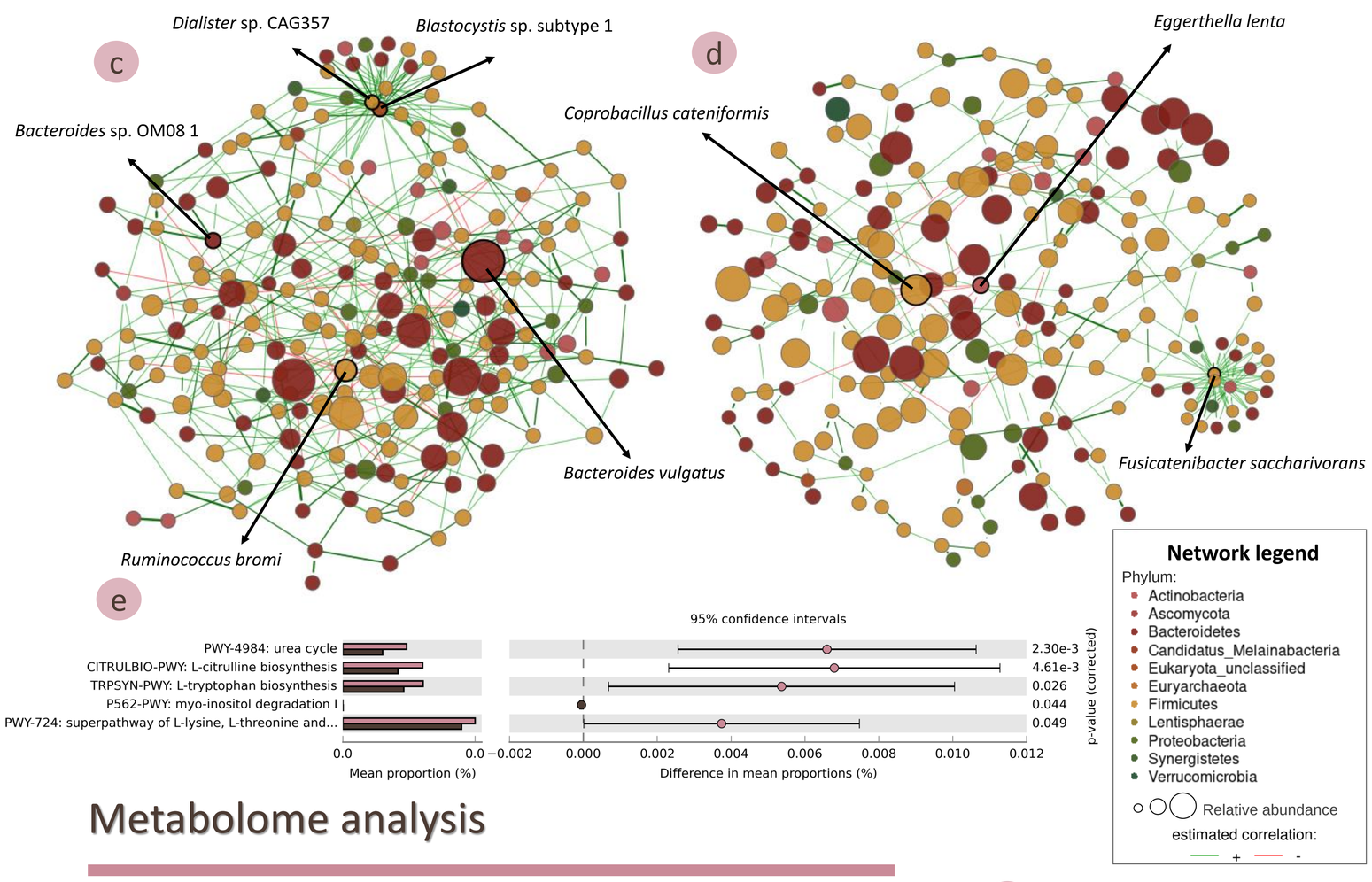
Celiac Disease (CeD) is a chronic immune-related inflammatory disease that affects the small bowel[1]. The gut microbiome (GM) is one of the non-gluten environmental factors contributing to CeD development[2] and differs among subgroups of CeD patients showing different clinical manifestations[3].
Data integration enables an understanding of the microbiome's role in health and disease from a more comprehensive perspective.
We aim to study the association of microbiome taxonomic profiles with the persistence of symptoms in non-responsive Celiac Disease patients (NR-CeD) -a pathology characterised by the persistence of symptoms from CeD after one year on a gluten-free diet[4]- from an integrative perspective.



Data analysis

After Multiple Factor analysis, we selected the most informative clinical variables: biochemistry, gluten free diet compliance, anthropometrics, symptoms persistence.
Afterward, we used model-based clustering over the Canberra distance of the selected variables (Fig. a), and we found two clusters of patients characterized by differences in symptoms prevalence but similar beta and alpha microbiome diversities (Fig. b).





Microbial community analysis

The microbial community structure analysis revealed differences according the predominant keystone taxa, network properties (dissimilarity, path length, edge Fig. c,d) and the microbial metabolic pathways (Fig. e).

- **Cluster 1 "Low-grade symptoms" (Fig. c):** Was enriched in SCFA producer's bacteria, and the eukaryotic *Blastocystis* sp., related to higher microbial richness and key in maintaining host immune homeostasis[8]. Also contain metabolic pathways related to amino acid biosynthesis and nitrogen metabolism pathways.
- **Cluster 2 "High-grade symptoms" (Fig. d):** Characterised by the presence of *Eggerthella lenta, Coprobacillus cateniformis,* and *Fusicatenibacter saccharivorans,* microorganisms related to modulation of immune system trough Th17 cells and IL-10 production[9,10].

Metabolome analysis

We found 372 metabolites in the faecal metabolome and were annotated according to the biological disposition (Fig g.) Differential analysis revealed that patients from Cluster 2 showed higher levels of the microbial-derived metabolites Indoxyl sulfate, Indolelactic acid -related with Tryptophan metabolism- cadaverine indicative of protein hydrolysis, and 2,6-Diaminopimelic acid, a key component of the bacterial cell wall that associated with enzymatic breakdown of gram-negative gut microbes.

Conclusions & Future perspectives

Data integration allows us to identify two clusters of patients with NR-CeD with microbial and functional differences in the microbiome, characterised by a proteolytic pro-inflammatory microbiome and higher intestinal and extraintestinal symptoms prevalence in Cluster 2 patients.

These results set the basis for future research regarding NR-CeD associations focusing on studying specific taxa and markers.

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